

Journal Club 19.02.2026

XPS/NIXSW of QA on Ag(100)

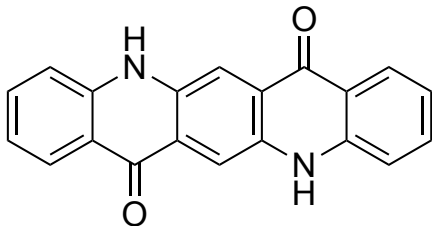
Lucas Hirschfeld

February 19, 2026

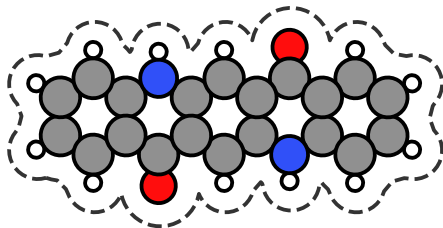
Agenda

- 1 Previous studies
- 2 Experimental/Theory
- 3 XPS of QA on Ag(100)
- 4 NIXSW of QA on Ag(100)
- 5 Structure models of QA on Ag(100)

Quinacridone (QA)



(a) Skeletal structure



(b) Space-filling model

α -phase of QA on Ag(100)¹

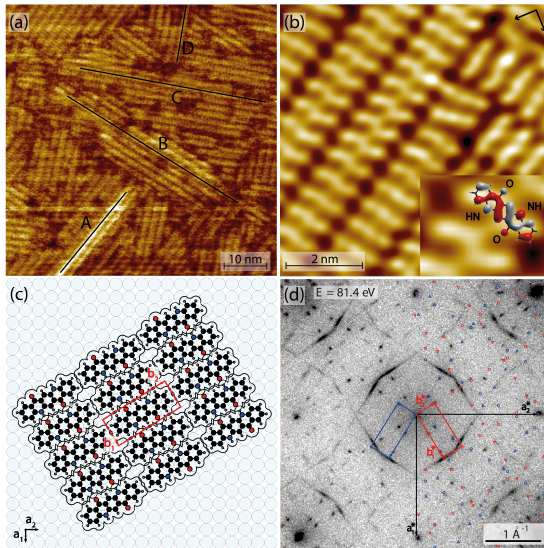


Figure: Literature overview of the α -phase of QA on Ag(100).

β -phase of QA on Ag(100)¹

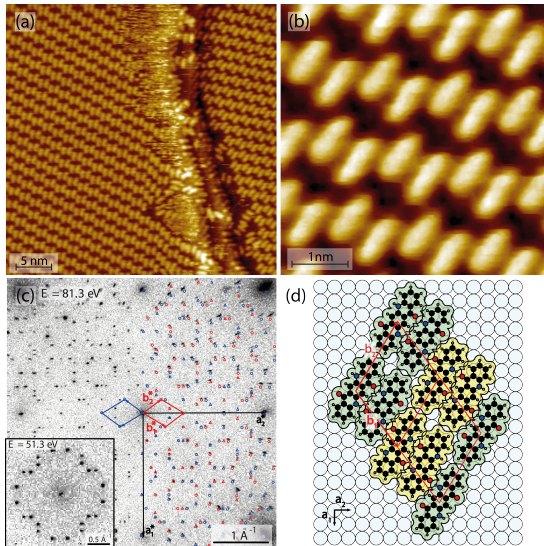


Figure: Literature overview of the β -phase of QA on Ag(100).

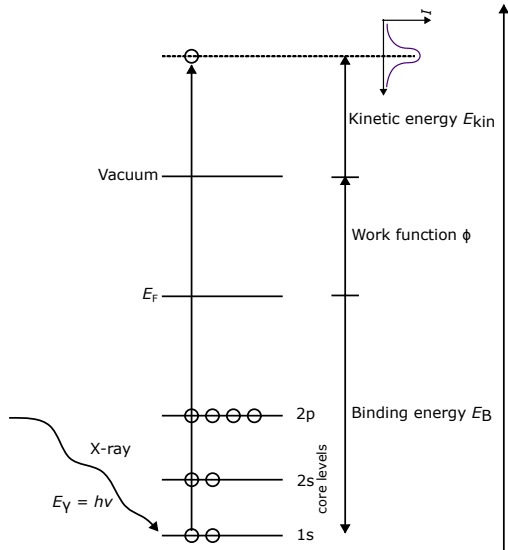


Figure: Schematic illustration of the XPS process.

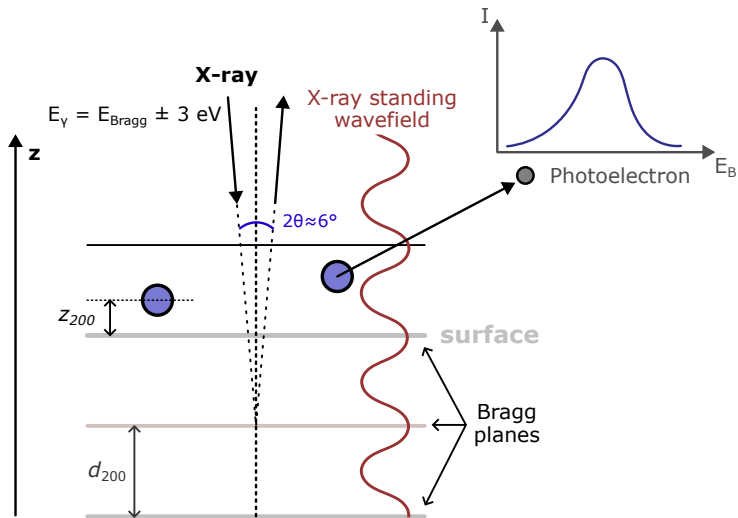
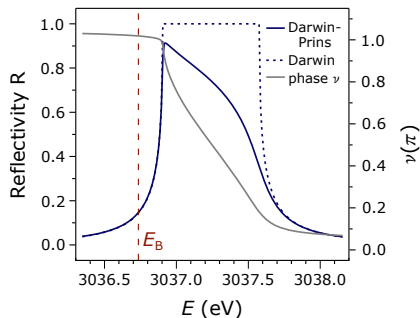
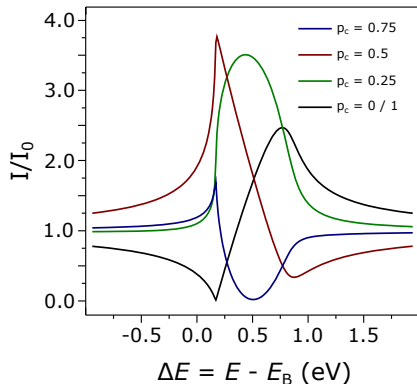


Figure: Schematic illustration of the NIXSW technique.



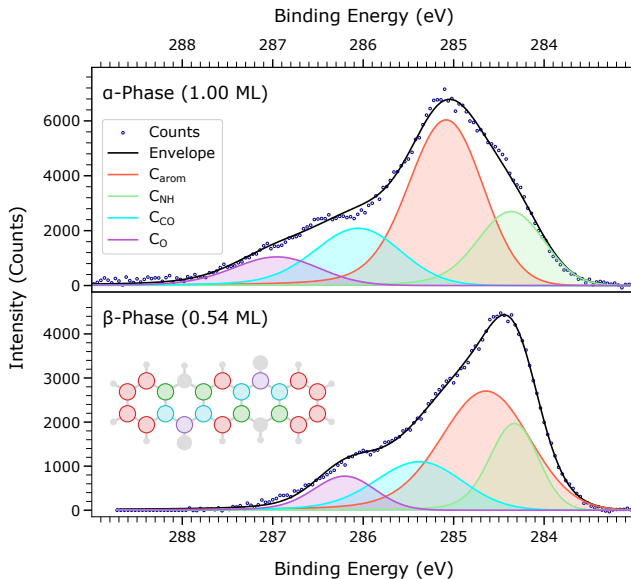
(a) Reflectivity



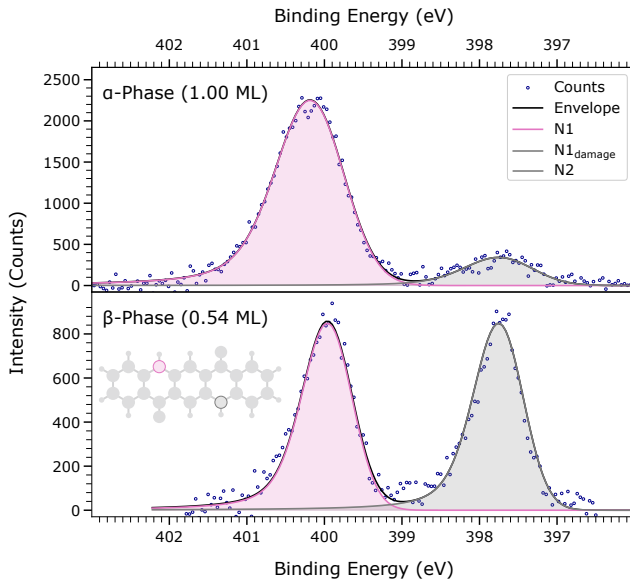
(b) Yield

$$\frac{I}{I_0} = 1 + S_R R(E_\gamma) + 2|S_I| \sqrt{R(E_\gamma)} \underbrace{f_c}_{\text{Darwin-Prins}} \cos(\nu(E_\gamma) - 2\pi \underbrace{p_c}_{\text{Darwin}} + \psi) \quad (1)$$

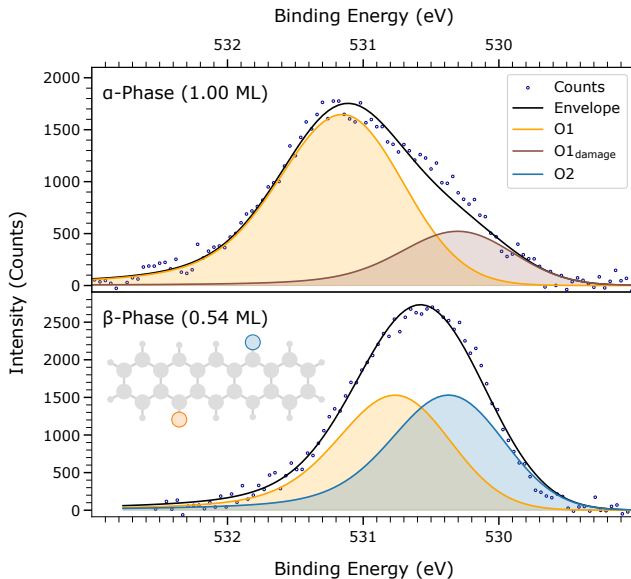
XPS: C1s



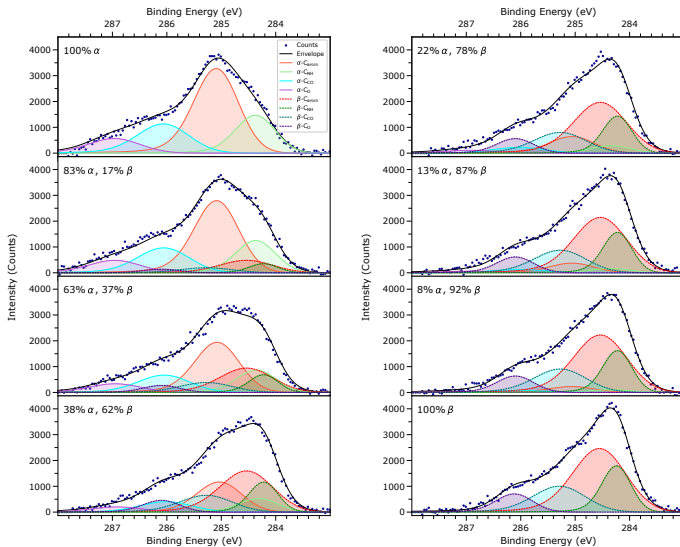
XPS: N1s



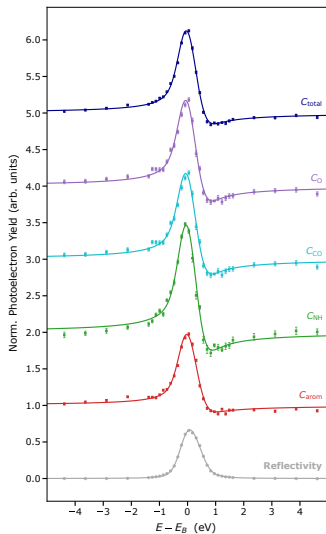
XPS: O1s



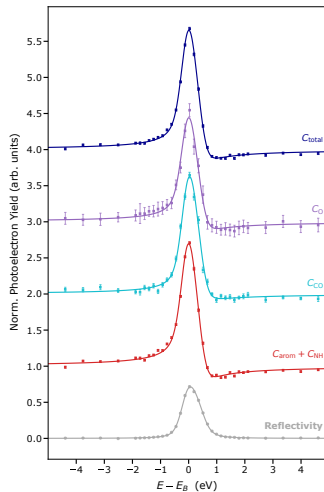
XPS: C1s Phase transition



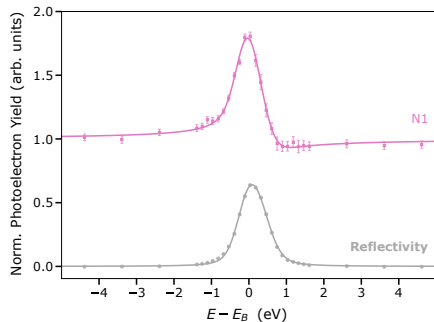
NIXSW: C1s



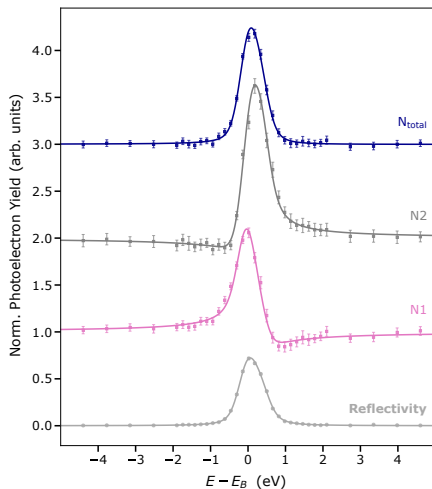
(a) α -phase



(b) β -phase

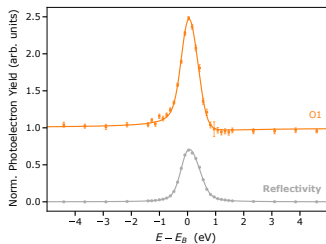


(a) α -phase

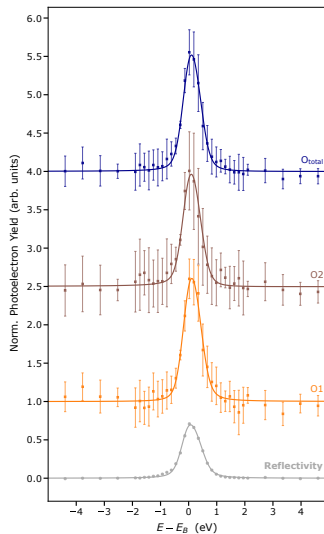


(b) β -phase

NIXSW: O1s

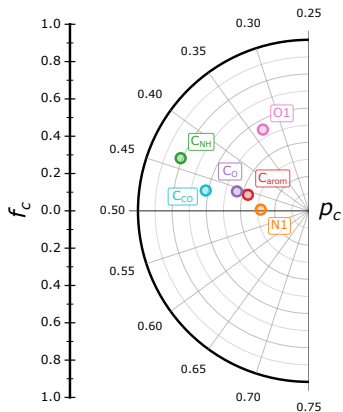


(a) α -phase

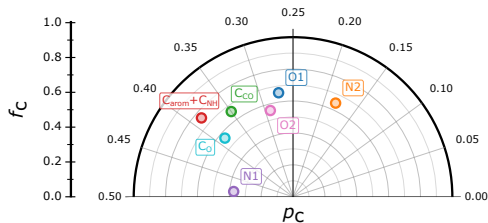


(b) β -phase

NIXSW: Comparison of the two phases

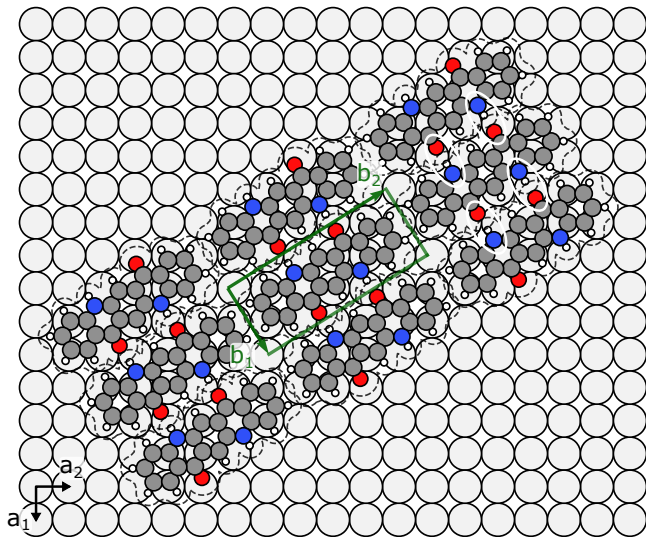


(a) α -phase

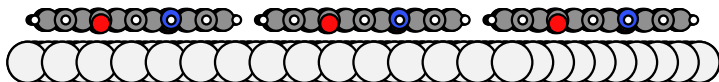


(b) β -phase

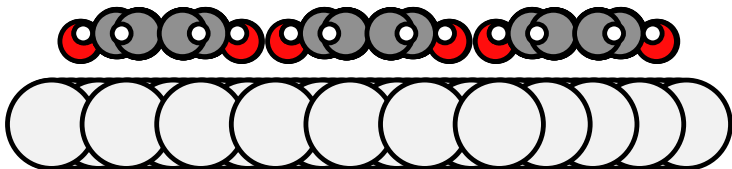
α -phase of QA on Ag(100)



α -phase of QA on Ag(100)

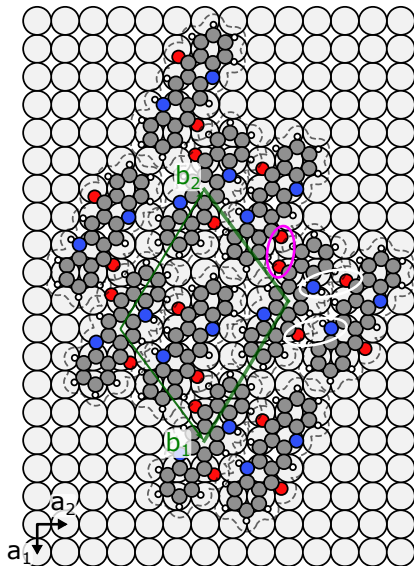


(a) Long side

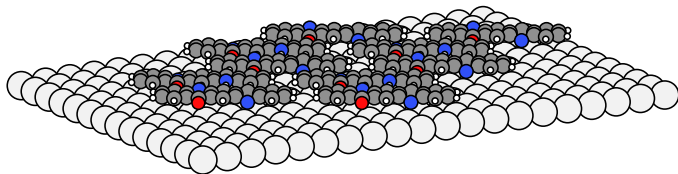


(b) Short side

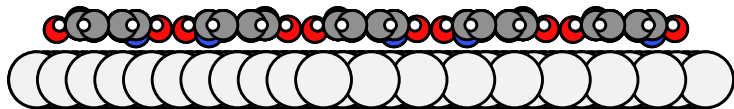
β -phase of QA on Ag(100): Homochiral



β -phase of QA on Ag(100): Homochiral

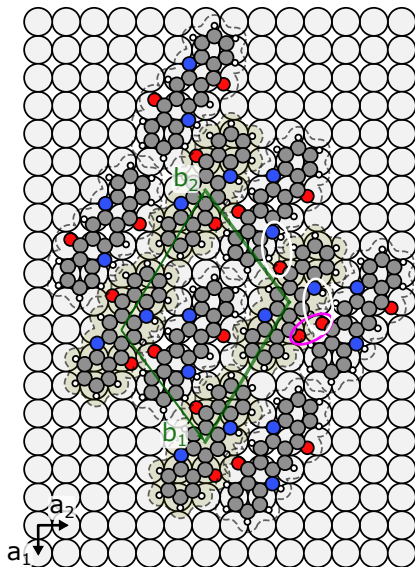


(a) Long side

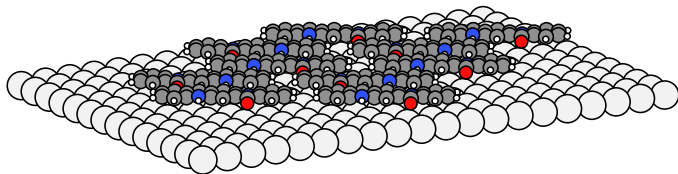


(b) Short side

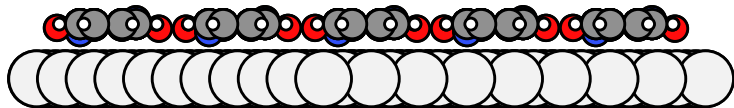
β -phase of QA on Ag(100): Heterochiral



β -phase of QA on Ag(100): Heterochiral

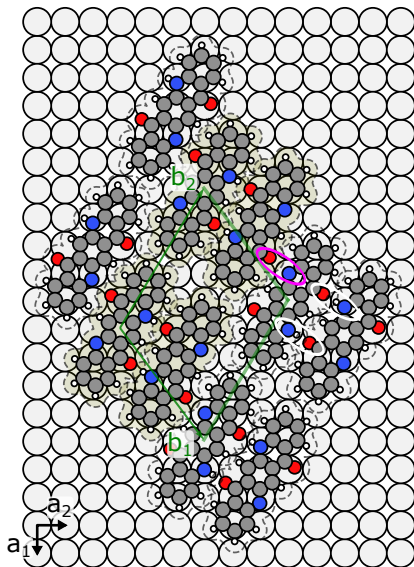


(a) Long side

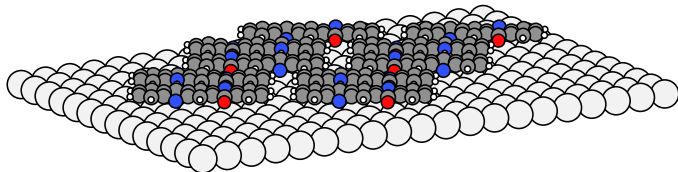


(b) Short side

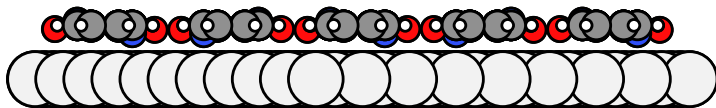
β -phase of QA on Ag(100): Dimers



β -phase of QA on Ag(100): Dimers



(a) Long side



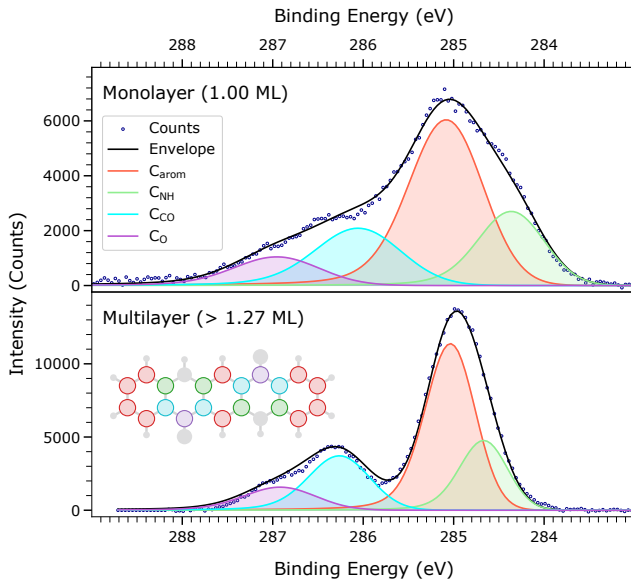
(b) Short side

- [1] [Hans Niklas Humberg](#). "One-Dimensional Aggregates of the Organic Dye Quinacridone on Metallic and Dielectric Surfaces." Thesis. Universitäts- und Landesbibliothek Bonn, July 21, 2025.

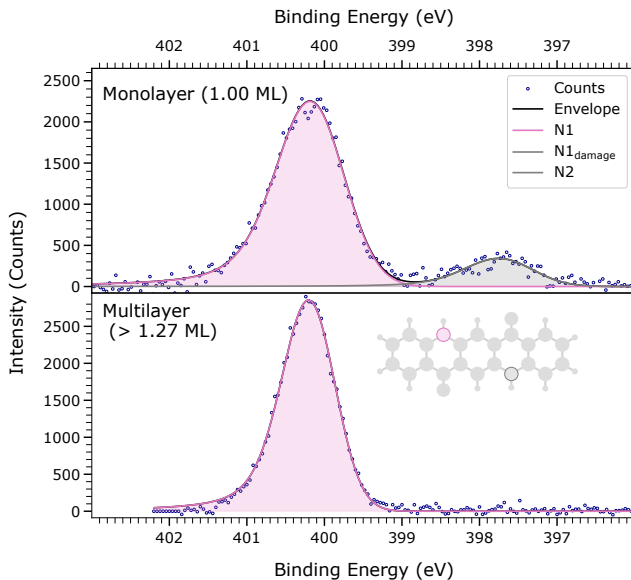
Thank you

Questions?

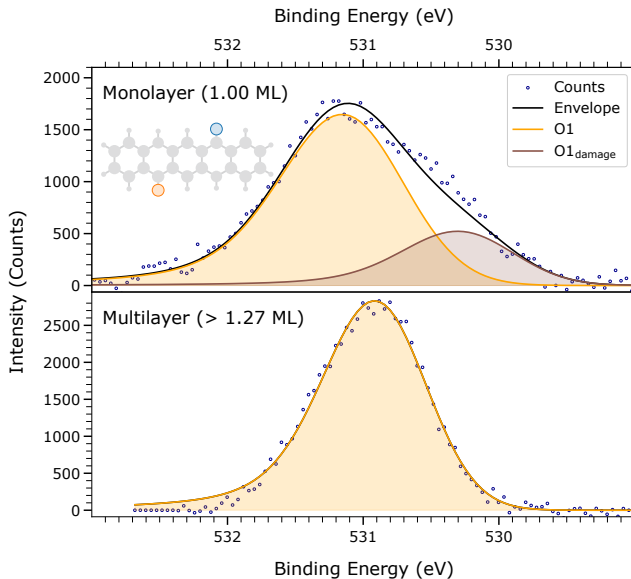
XPS: C1s Multilayer



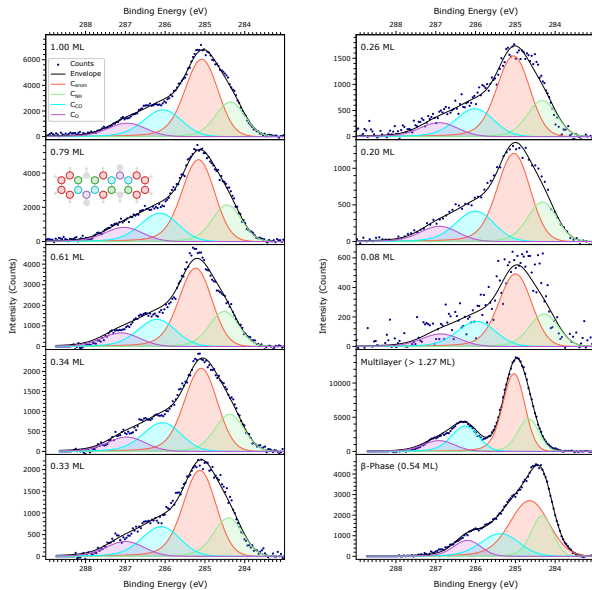
XPS: N1s Multilayer



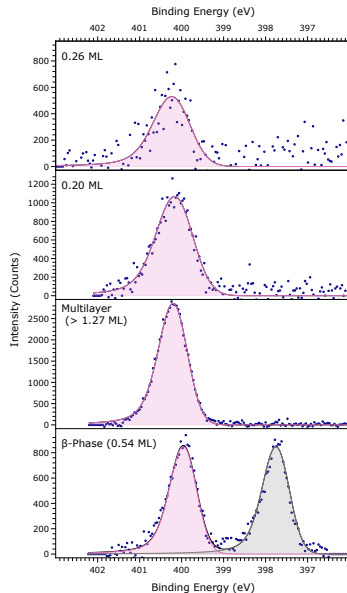
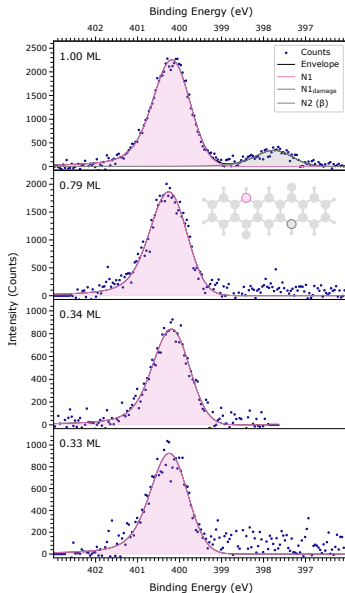
XPS: O1s Multilayer



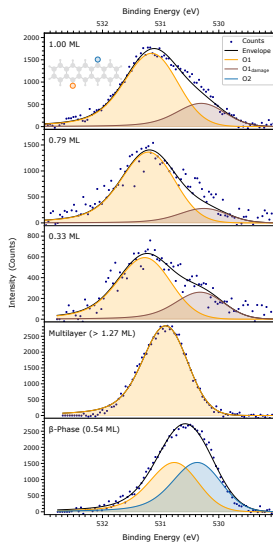
XPS C1s



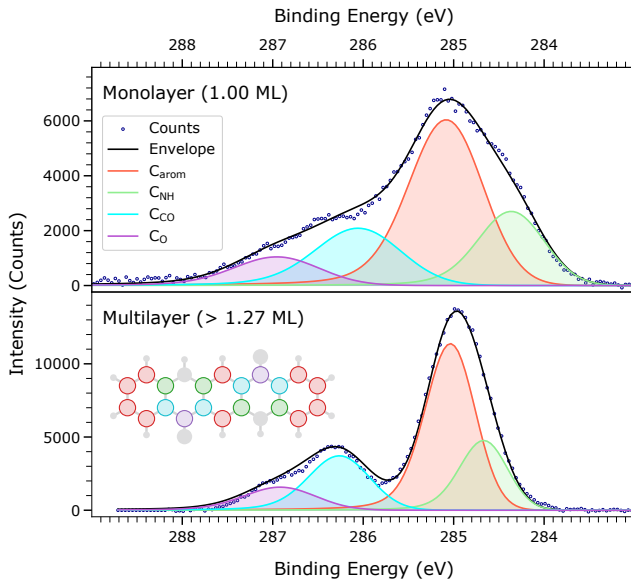
XPS N1s



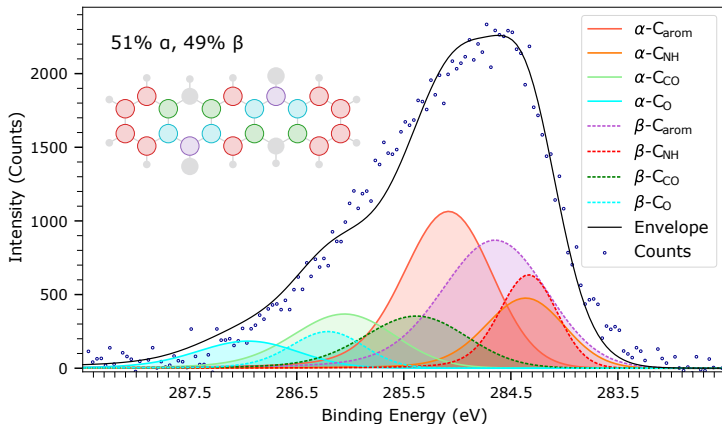
XPS O1s



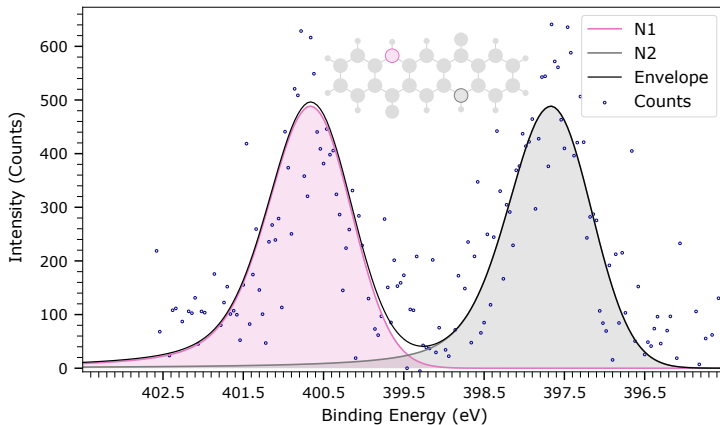
XPS: C1s Multilayer



XPS: C1s Burnt Phase



XPS: N1s Burnt Phase



NIXSW Fit Model 3 Peaks

